SOLARGUARD: SMART MONITORING AND FORECASTING FOR ENERGY GRIDS

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TABLE OF CONTENTS

Approach & Methodology

1. Data

The data used in this project was collected from [site name], based on the technical specifications of a solar power plant designed for Qom province, Iran. The dataset contains a full year of hourly information for 2019, comprising 8,760 rows and 6 columns.

- time: This column contains the exact time of data entry, often in a standardized format like UTC.
- local_time: The timestamp for each data entry, which allows for the analysis of seasonal and daily trends.
- electricity: The project's main target variable; the amount of electricity generated by the power plant per hour. Our model's goal is to accurately predict this column.
- irradiance_direct and irradiance_diffuse: Two key columns that measure solar radiation. This data helps us precisely analyze the effect of sunlight on energy production.
- temperature: The ambient temperature for each hour. This is an essential column for evaluating the impact of temperature on the efficiency of solar panels.

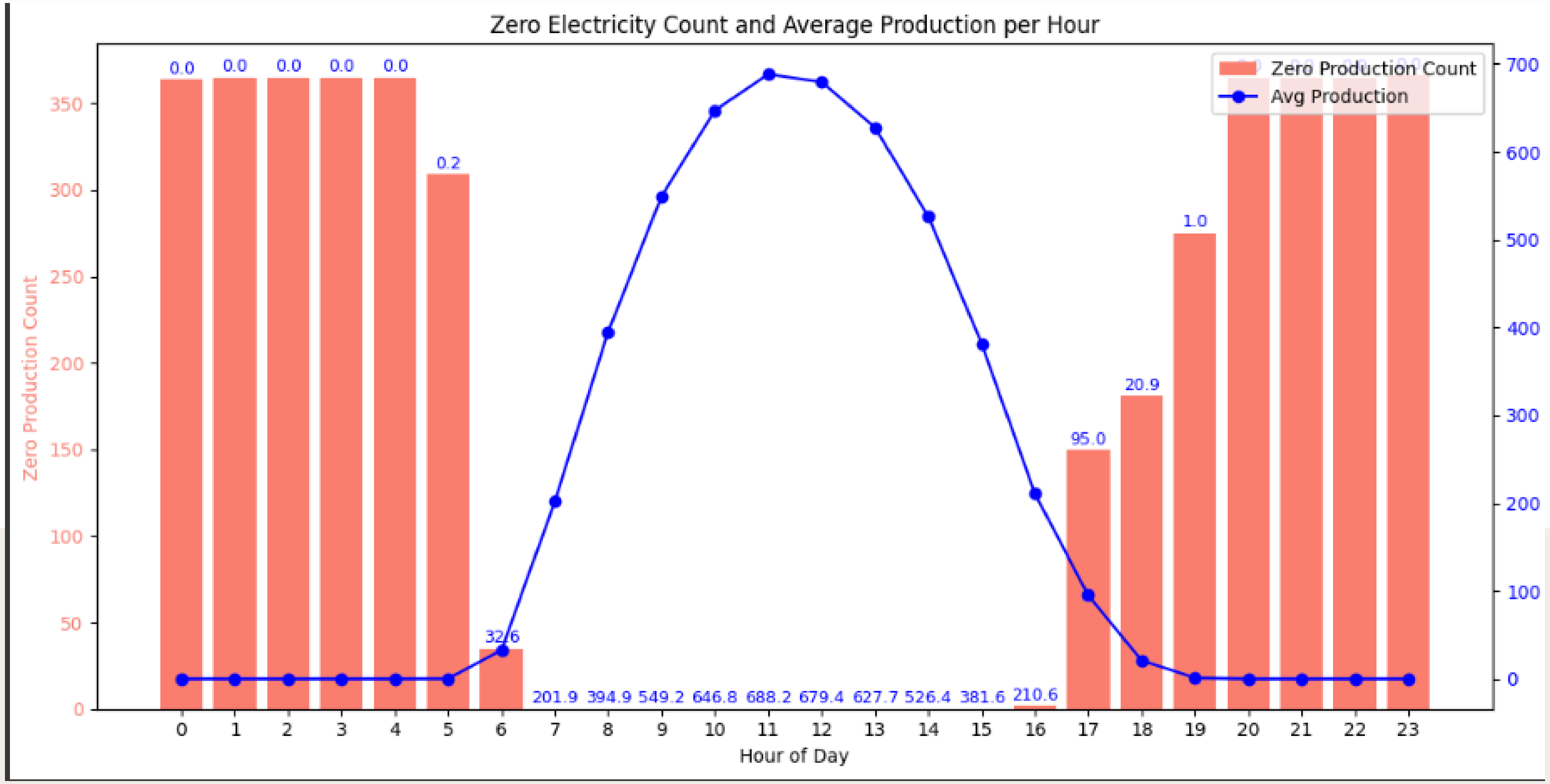
2. Data Preprocessing

Time as Index: We set the local_time column as the DataFrame index. This is crucial for time-series modeling, as it allows us to analyze the data's temporal frequency and structure.

Feature Engineering: We extracted statistical and time-domain features from the dataset. This includes creating new features that capture temporal patterns, which are vital for improving model performance.

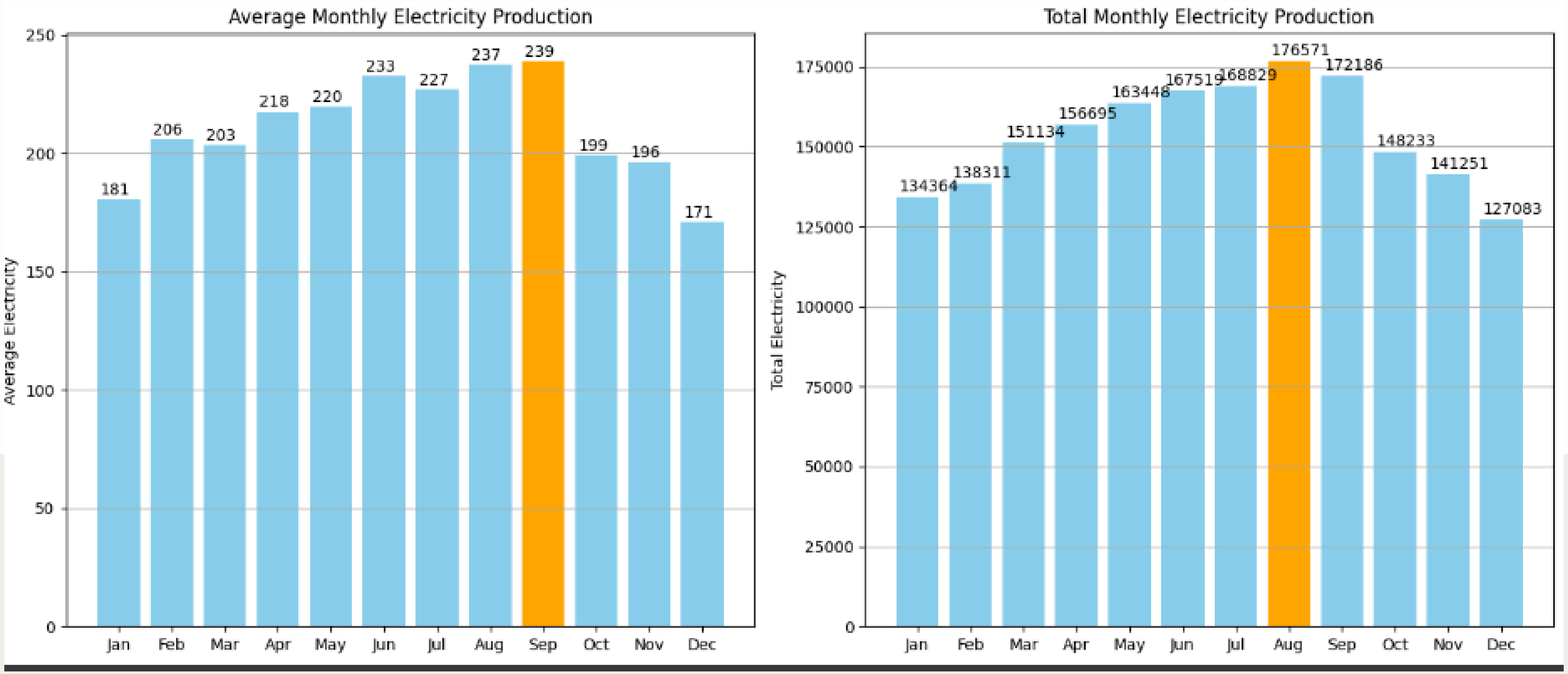
Outlier Detection: We identified and handled outliers within the data. Addressing these extreme values is an important step to ensure the model's accuracy and robustness.

Hours Without Electricity Production and Average Daily Electricity Output



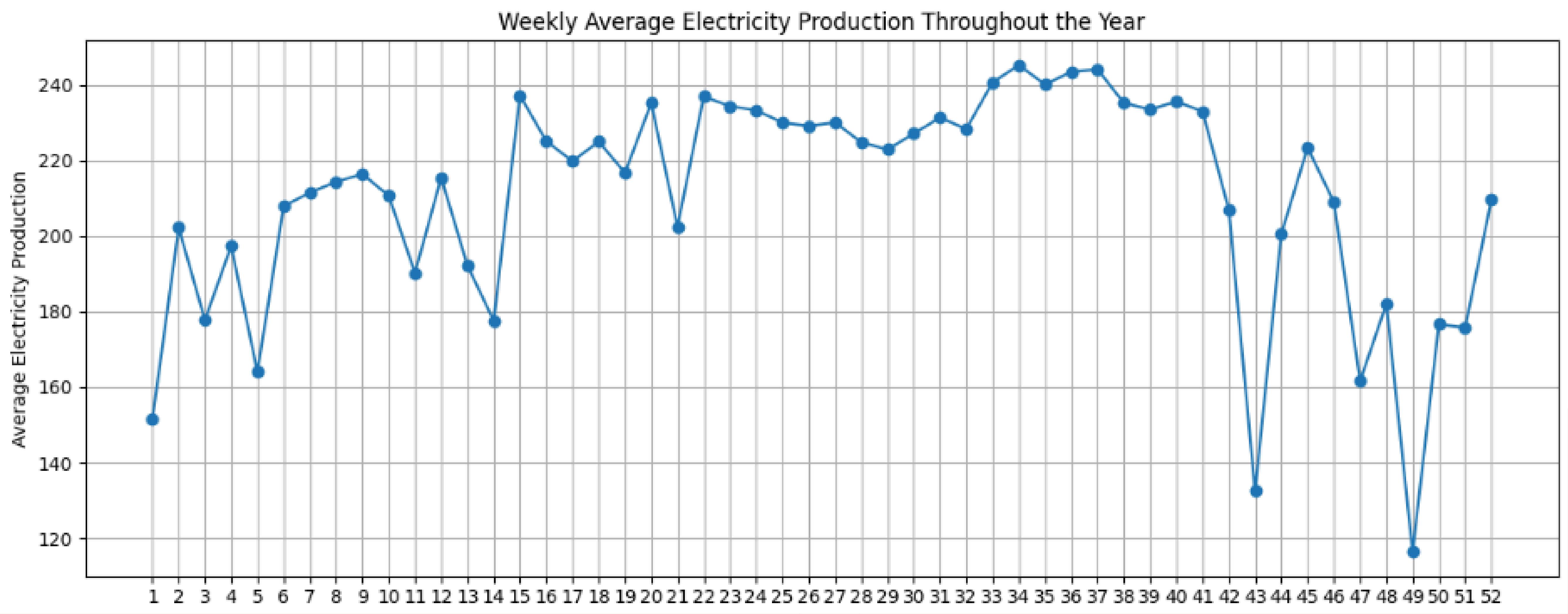
This chart simultaneously shows the number of times electricity production was zero for each hour of the day (in red bars) and the average electricity production for the same hour (in a blue line). This information helps identify the hours with a high probability of zero production and highlights daily electricity generation patterns.

Comparison of Monthly Average and Total Electricity Production



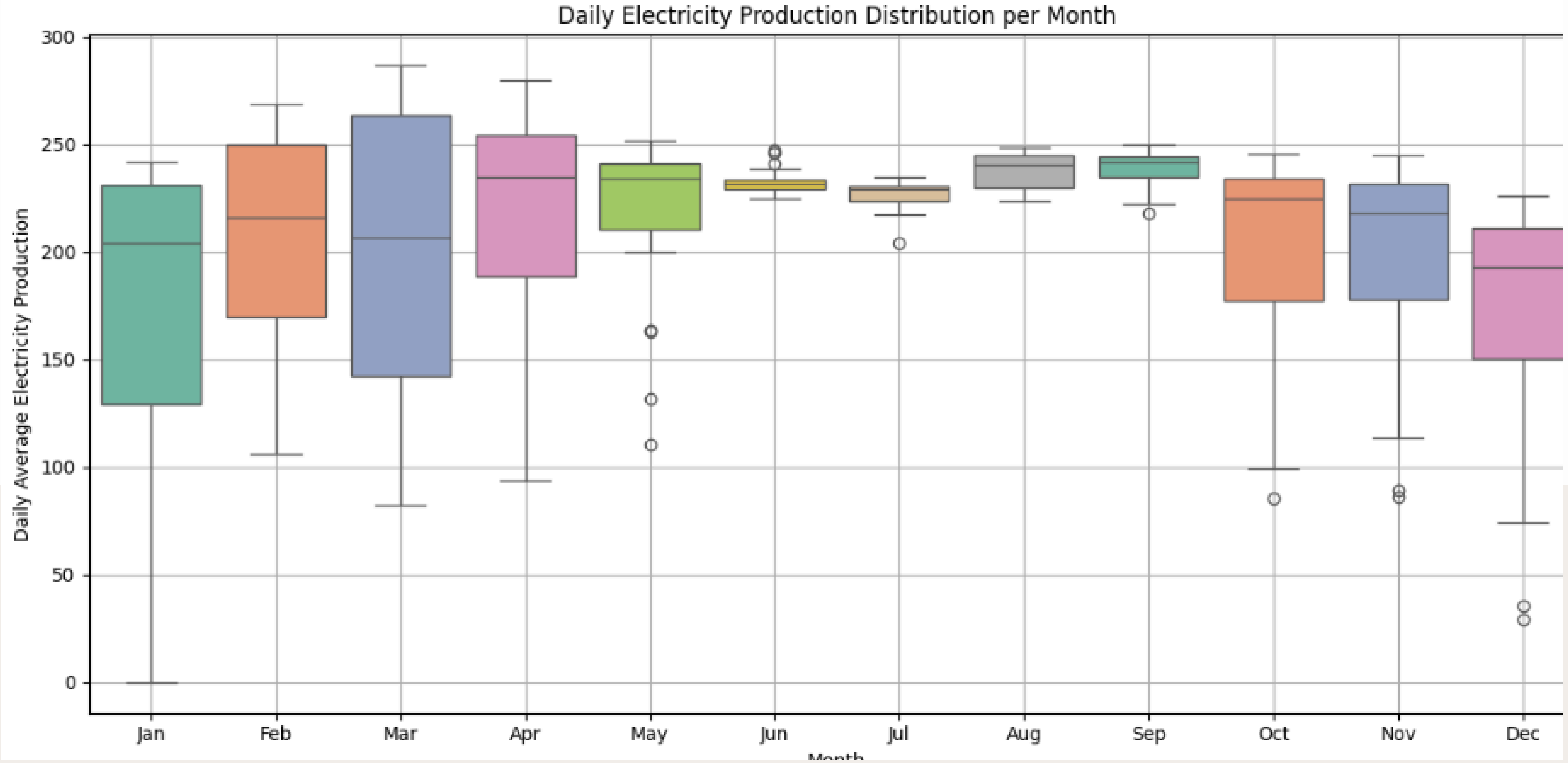
The average production chart shows the plant's daily efficiency patterns throughout the year, while the total production chart indicates the overall monthly output and revenue for financial planning. Together, these two charts provide a complete picture of the solar plant's performance and its seasonal profitability.

Weekly Average Electricity Production Throughout the Year



This plot displays the average electricity production on a weekly basis throughout a year. By examining this chart, one can observe seasonal patterns and variations in electricity production across different weeks of the year, identifying periods of peak or reduced output.

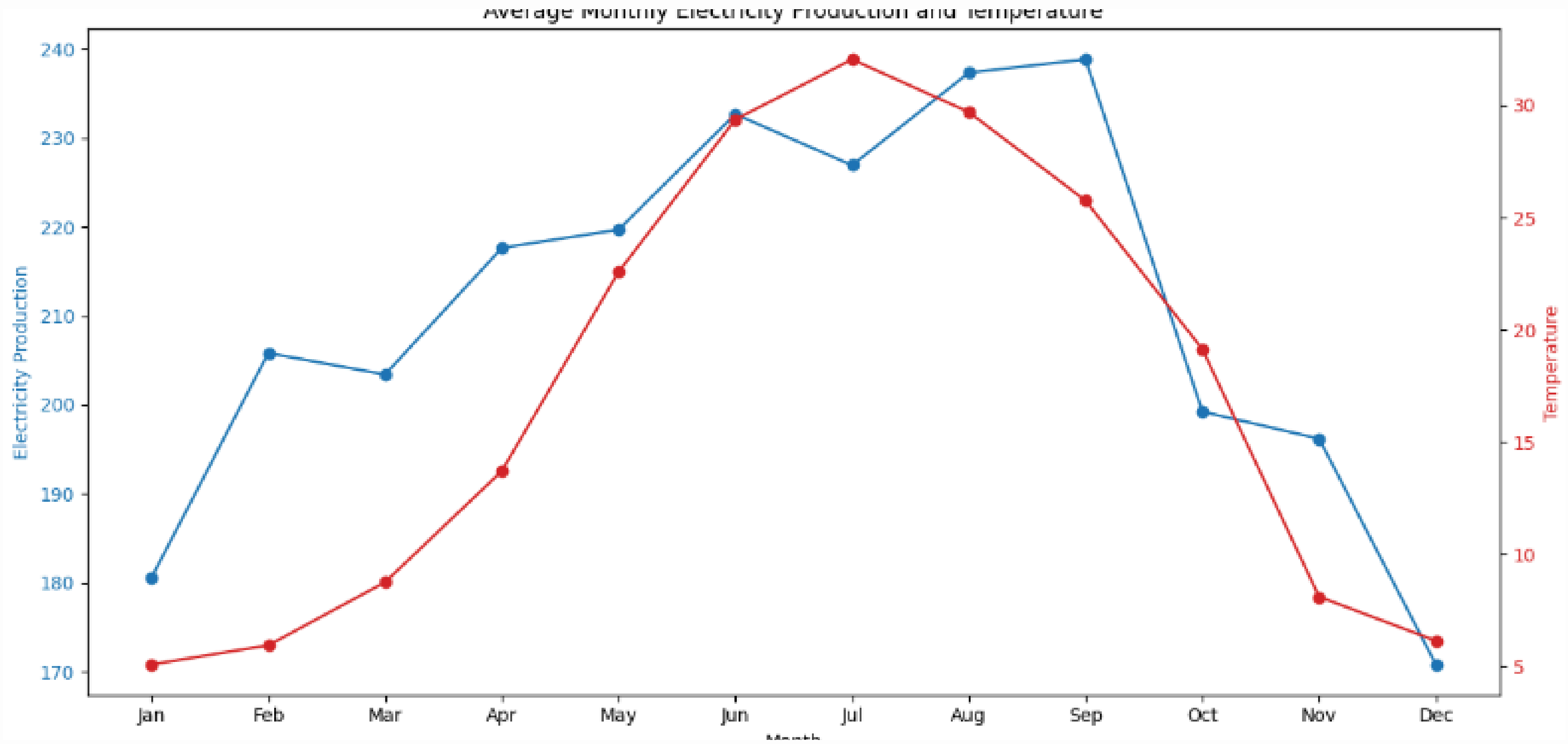
Daily Electricity Production Volatility per Month



The box plot analysis reveals a clear pattern of seasonal variability in electricity production:

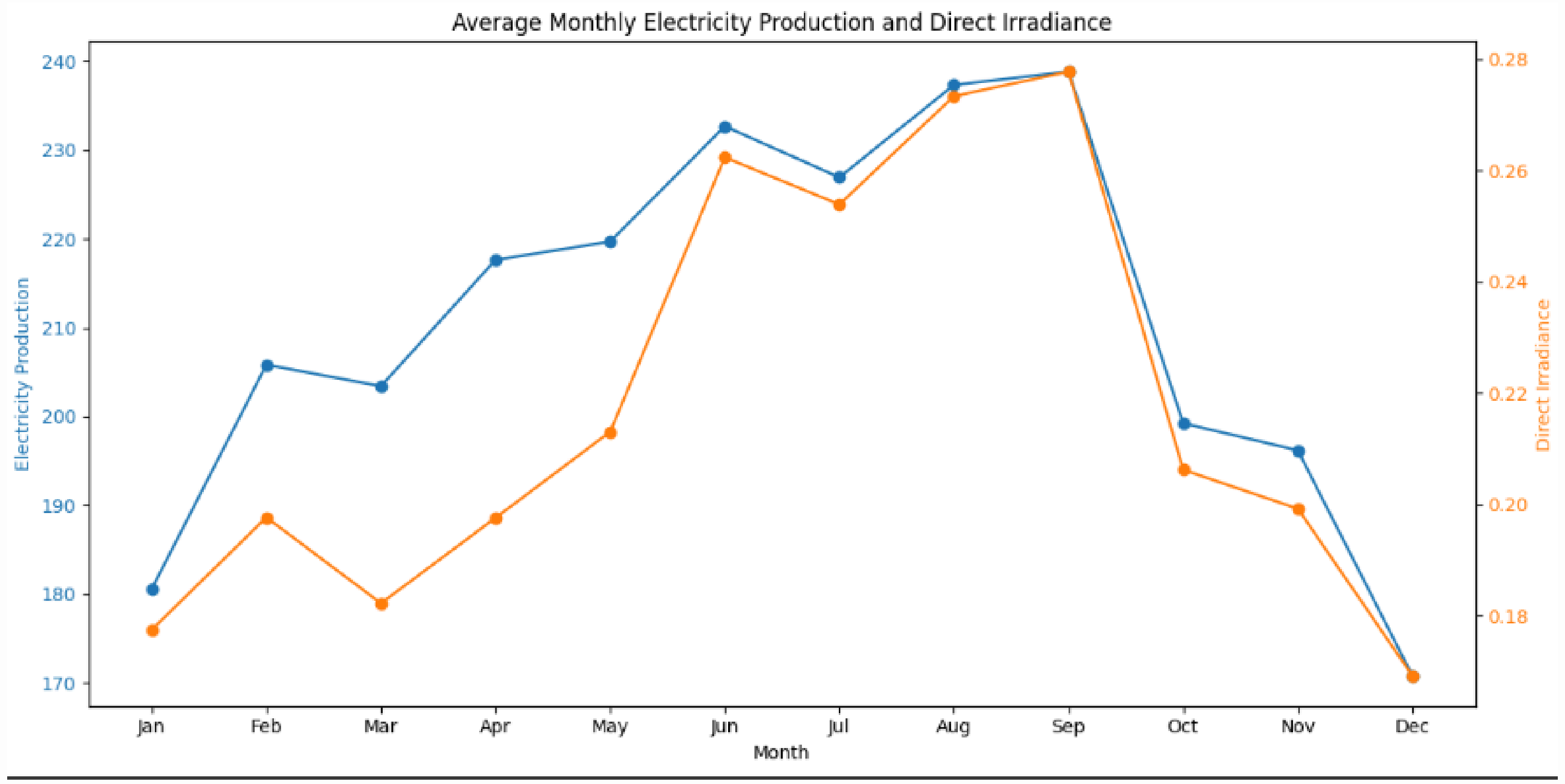
- Low Volatility in Summer: The months of June, July, August, and September show minimal fluctuation in daily output. The compact boxes and fewer outliers indicate stable and predictable weather conditions, leading to consistent power generation.
- High Volatility in Winter and Spring: In contrast, March, October, November, and December exhibit significant daily volatility. The longer boxes and numerous outliers suggest that sudden weather changes—such as cloudy, rainy, or stormy days—cause a sharp drop in electricity production, making it less stable and predictable during these seasons.

Correlation Between Electricity Production and Temperature



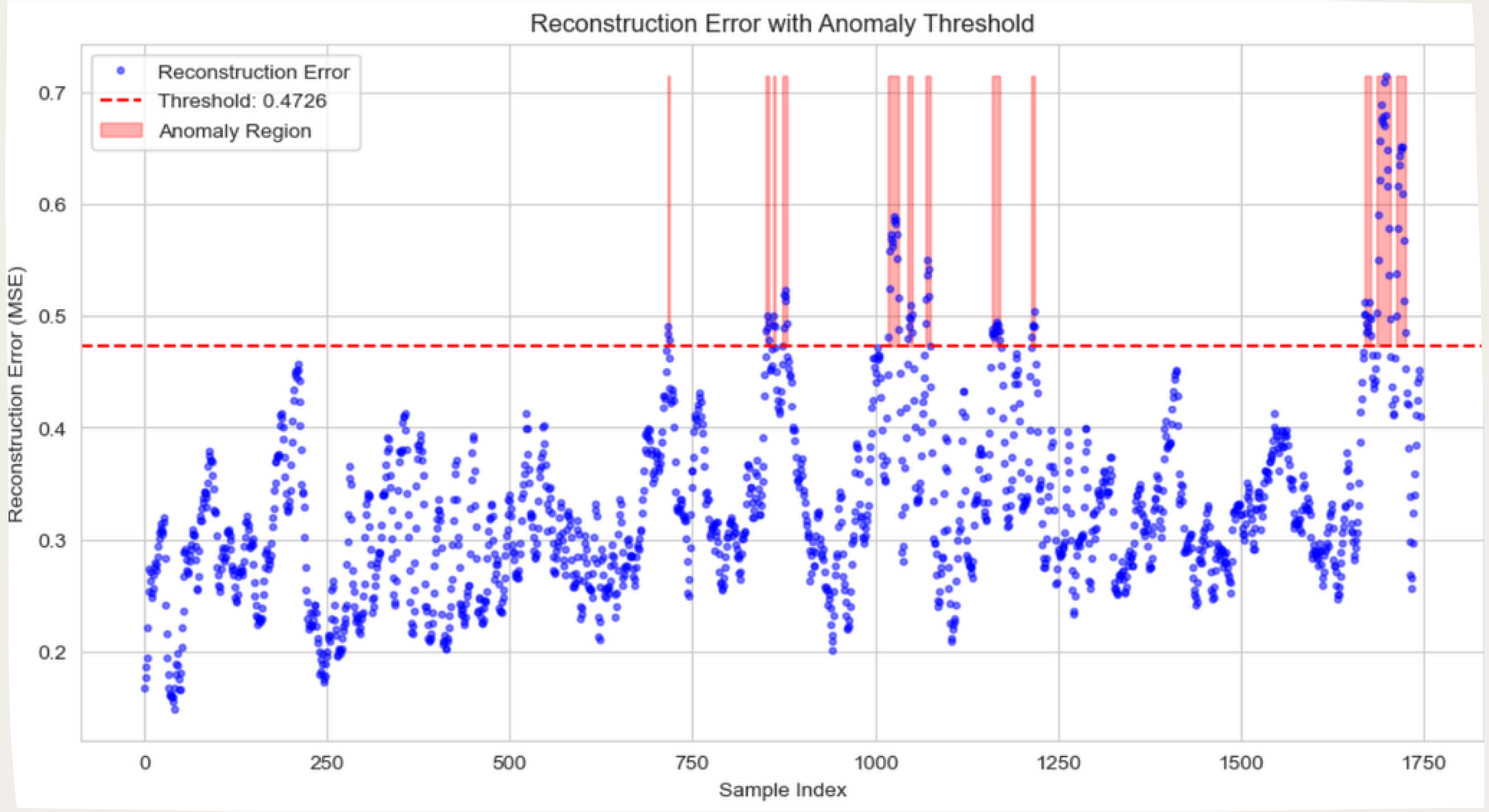
This chart shows a strong positive correlation between electricity production and temperature for most of the year. Both the blue line (electricity production) and the red line (temperature) rise and peak together. However, a key observation is that in July, as the temperature peaks, electricity production slightly dips, and the peak production shifts to September. This suggests that excessively high temperatures can have a negative impact on the efficiency of solar panels.

Correlation Between Electricity Production and Solar Irradiance



This chart clearly shows a strong positive correlation between electricity production and direct solar irradiance. Both the blue line (electricity production) and the orange line (irradiance) follow nearly identical trends throughout the year, peaking in the warmer months and dropping in the colder ones. This confirms that solar irradiance is the primary factor in determining the plant's electricity output.

Autoencoder Intelligent Anomaly Detection: Monitoring Energy Production Health with LSTM Autoencoder

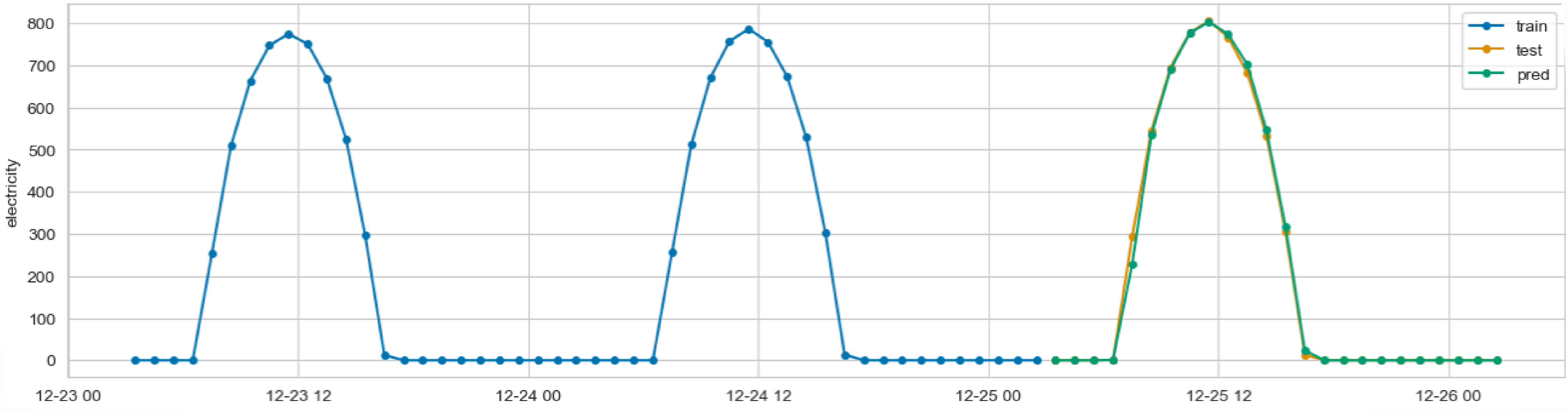


Model: We utilize a Recurrent Neural Network (LSTM Autoencoder) trained exclusively on normal summer/spring patterns. The model learns to compress and precisely reconstruct thousands of multivariate data sequences (including electricity, temperature, and irradiance).

Detection Mechanism: Anomalies are identified based on the Reconstruction Error (RE). If the model fails to accurately reconstruct a 24-hour sequence (as seen in the highlighted regions of the graph), it flags the event as a critical anomaly or grid disturbance.

Crisis Contribution: This model functions as an Early Warning System, detecting subtle deviations before they escalate into widespread outages. This precision enables Proactive Maintenance, securing grid stability and optimizing energy resources during times of crisis.

Forecasting Energy Production: A Sktime Reduction Approach



Function: It successfully captures the highly non-linear, sinusoidal daily production cycle (train, test, and prediction lines align closely).

Accuracy (MAE): The model demonstrates high fidelity in predicting the peak generation periods but exhibits a high Mean Absolute Error (MAE: 225.95), indicating accumulated error during the low-production hours and transition points (dawn/dusk).

Conclusion: The model provides a reliable 24-hour forecast of energy generation, establishing a strong foundation for optimizing energy storage and grid management, though it requires further feature engineering (Sin/Cos features) to minimize edge error.